

1- Report default training hyperparameters for each model

For each model (CSBrain, CBraMod, EEGMamba, Codebrain, Biot, LaBraM), open the main finetuning script (finetune_main.py or equivalent) and extract the default values for:

- Learning rate (--lr)
- Weight decay (--weight_decay)
- Batch size (--batch_size)
- Number of epochs (--epochs)
- Optimizer (--optimizer)
- Warmup epochs/steps (if any)
- LR scheduler (cosine, linear, etc.)
- Gradient clipping (--clip_value or equivalent)
- Label smoothing (--label_smoothing)
- Dropout (--dropout)
- Frozen backbone (--frozen)

Compile them into a single comparison table with one column per model so differences are immediately visible.

	LABRAM	EEGPT	BIOT	CBRAMOD	CSBRAIN	CODEBRAIN
LR	5e-4	5e-4	1e-3	1e-4	1e-4	1e-4
WD	0.05	0.05	1e-5	5e-2	5e-2	5e-4
BS	64	64	512	64	64	64
EPOCH	30	30	100	50	50	50
OPT	adamw	adamw	Adam	AdamW	AdamW	AdamW
WARMUP EPOCH	5	5	–	–	–	–
LR SDL	cosine	cosine	?	–	cosine	–
CLIP VALUE	None	none	–	1	1	5
SMOOTH	0.1	0.1	–	0.1	0.1	0.1
DROPOUT	0.0	0.0	0.2?	0.1	0.1	0.1
FROZEN	–	–	–	False	False	False

#cosine=cosine_scheduler

/EEG-FM/Biot/run_binary_supervised.py

```

elif args.model == "CNNTransformer":
    model = CNNTransformer(
        in_channels=args.in_channels,
        n_classes=args.n_classes,
        fft=args.sampling_rate,
        steps=args.hop_length // 5,
        dropout=0.2,
        nhead=4,
        emb_size=256,
    )

```

```
return result
```

```
def configure_optimizers(self):  
    optimizer = torch.optim.Adam(  
        self.model.parameters(),  
        lr=self.args.lr,  
        weight_decay=self.args.weight_decay,  
    )  
  
    return [optimizer] # , [scheduler]
```